

Subject Name: Data Preprocessing and EDA

Subject Code (Credits): CUGG1004 (2+2+0)

Teaching and Evaluation Scheme

Teaching Scheme				Evaluation Scheme				
T	PR	PJ	Total Credits	Theory External	Theory Internal	Practical External	Practical Internal	Total Marks
Hrs	Hrs	Hrs		Marks	Marks	Marks	Marks	Marks
2	2	0	4	70	30	50	50	200

Outline of the Course:

Sl. No	Module Name	Minimum Hours
1	Preprocessing Basics	18
2	Feature Engineering	18
3	Data Visualization	12
4	Case Studies & Tools	18
5	Advanced Data Handling	18

Total Hours (Theory): 42

Total Hours (Lab Practice): 42

Total Course Hours: 84

Detailed Breakup of the Course:

Module No.	Topics in the Module	No. of Lecture Hours
1	Preprocessing Basics: Data Preprocessing Basics: raw data formats, data cleaning techniques; Handling Missing & Categorical Data: imputation methods, one-hot and label encoding; Outliers & Scaling: outlier detection and removal, scaling and normalization techniques; Practice: Load CSV/JSON files, handle missing values, encode categorical data, detect/remove outliers, apply Min-Max normalization and StandardScaler.	18
2	Feature Engineering: Feature Selection: correlation methods, statistical tests; Dimensionality Reduction: PCA, t-SNE techniques; Feature Creation: interaction terms, polynomial features, importance evaluation; Practice: correlation matrix, chi-square and ANOVA feature selection, PCA, t-SNE visualization, polynomial feature generation.	18
3	Data Visualization: Visualization Tools: Matplotlib, Seaborn libraries; Plot Types: bar, line, scatter, histograms, heatmaps; Interactive & Insights: Plotly visualizations, correlation analysis, decision-making; Practice: line/bar/scatter plots, histogram/box plots, correlation heatmap, interactive Plotly dashboard, multi-plot layout	12
4	Case Studies & Tools: Pipelines & Tools: Scikit-learn pipelines, AutoML introduction; Data Analysis: real-world cleaning, open dataset exploration; Projects: group-based exploratory data analysis tasks; Practice: build sklearn pipeline, clean messy dataset, pandas-profiling, AutoML with PyCaret, end-to-end EDA .	18
5	Advanced Data Handling: Time & Text Data: time series preprocessing, text cleaning (tokenization, stop-word removal, stemming, lemmatization); Imbalanced & Augmentation: oversampling/undersampling (SMOTE, ADASYN), data augmentation; Pipelines & Practices: automation using Pandas, Dask, large-scale preprocessing best practices; Practice: time-series resampling, text cleaning, SMOTE/ADASYN, text augmentation, automated preprocessing pipeline.	18
Total Number of Hours		84

COs Code	Course Outcome	BT Level
CO1	Apply data preprocessing techniques to clean and prepare raw datasets for analysis.	Remember, Understand

CO2	Analyze data features and implement feature engineering techniques to improve model performance.	Apply
CO3	Develop data visualization techniques to uncover patterns and insights from data.	Apply, Analyze, Evaluate
CO4	Evaluate preprocessing pipelines for diverse datasets using automation tools.	Apply, Analyze, Create
CO5	Design workflows for handling time-series, text, and imbalanced datasets effectively	Apply, Analyze
CO6	Construct an end-to-end data preprocessing and EDA pipeline integrating all modules to solve a real-world data problem.	Apply, Analyze, Create

CO/PO/ PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	1	1	2	1	-	-	-	-	-	1	2	2	1
CO2	2	3	3	2	3	-	-	1	-	-	-	1	3	2	1
CO3	1	3	2	2	3	-	2	-	-	-	-	1	2	3	1
CO4	2	2	3	3	3	-	-	-	-	2	-	2	3	2	2
CO5	2	3	3	2	3	-	-	-	-	-	-	2	3	3	2
CO6	2	3	3	3	3	-	-	-	2	2	2	2	3	3	2

Practice marks criteria : 1 assignment per module. Total 5 assignments
10 marks each : 5-implementation and output . 5-clarity

Session Plan

Session 1

Topic: Raw data formats; data cleaning intro

Materials: <https://www.kaggle.com/learn/pandas> (Pandas intro, loading CSV/JSON, missing values)

Session 2

Topic: Missing data: imputation methods (mean, median, mode) with Pandas demo

Materials: <https://scikit-learn.org/stable/modules/impute.html> (Official scikit-learn imputation docs)

Session 3

Topic: Exp 1.1 + 1.2

Materials: —

Session 4

Topic: Categorical encoding: one-hot & label encoding

Materials: <https://scikit-learn.org/stable/modules/preprocessing.html> (Encoding, scaling, preprocessing)

Session 5

Topic: Outlier detection: Z-score & IQR theory + visualisation

Materials: <https://www.analyticsvidhya.com/blog/2021/05/feature-engineering-how-to-detect-and-remove-outliers-with-python-code/> (Outlier detection with Z-score & IQR)

Session 6

Topic: Exp 1.3 + 1.4

Materials: —

Session 7

Topic: Scaling & normalisation: Min-Max vs standardisation

Materials: <https://scikit-learn.org/stable/modules/preprocessing.html> (Same page covers MinMaxScaler and StandardScaler — reuse across Weeks 2–3)

Session 8

Topic: Module 1 review, Q&A, common real-world pitfalls

Materials: <https://scikit-learn.org/stable/modules/preprocessing.html> (Same page covers MinMaxScaler and StandardScaler — reuse across Weeks 2–3)

Session 9

Topic: Exp 1.5

Materials: —

Session 10

Topic: Feature selection: correlation methods overview

Materials: https://scikit-learn.org/stable/modules/feature_selection.html (Official feature selection docs — χ^2 , ANOVA, mutual info all covered)

Session 11

Topic: Statistical tests for feature selection: χ^2 , ANOVA, mutual info

Materials: https://scikit-learn.org/stable/modules/feature_selection.html (Official feature selection docs — χ^2 , ANOVA, mutual info all covered)

Session 12

Topic: Exp 2.1 + 2.2

Materials:

Session 13

Topic: Dimensionality reduction: PCA theory & intuition

Materials: <https://cs229.stanford.edu/notes2021fall/lecture14-pca.pdf> (Stanford CS229 PCA lecture slides — PDF)

<https://scikit-learn.org/stable/modules/decomposition.html> (PCA with scikit-learn)

Session 14

Topic: t-SNE: intuition, hyperparameters & use cases; compare with PCA

Materials: <https://scikit-learn.org/stable/modules/manifold.html> (t-SNE with scikit-learn)

Session 15

Topic: Exp 2.3 + 2.4

Materials: —

Session 16

Topic: Feature creation: polynomial features & interaction terms

Materials: <https://scikit-learn.org/stable/modules/preprocessing.html#polynomial-features> (Polynomial features)

Session 17

Topic: Feature importance evaluation: RF-based, permutation importance

Materials: https://scikit-learn.org/stable/modules/permutation_importance.html (Permutation importance & RF-based importance)

Session 18

Topic: Exp 2.5

Materials: —

Session 19

Topic: Matplotlib: line, bar, scatter — layout & styling principles

Materials: <https://matplotlib.org/stable/tutorials/index.html> (Official Matplotlib tutorials)

Session 20

Topic: Seaborn: histograms, box plots, pair plots; when to use each

Materials: <https://seaborn.pydata.org/tutorial.html> (Official Seaborn tutorial)

Session 21

Topic: Exp 3.1 + 3.2

Materials: —

Session 22

Topic: Heatmaps & correlation visualisation for EDA

Materials: <https://plotly.com/python/plotly-express/> (Plotly Express official docs & examples)

Session 23

Topic: Plotly: interactive dashboards; extracting decisions from visuals

Materials: <https://seaborn.pydata.org/generated/seaborn.heatmap.html> (Seaborn heatmap docs)

Session 24

Topic: Exp 3.3 + 3.4 + 3.5

Materials: —

Session 25

Topic: Scikit-learn pipelines: design, structure, transforms

Materials: <https://scikit-learn.org/stable/modules/compose.html> (Pipeline & ColumnTransformer official docs)

Session 26

Topic: AutoML intro: TPOT, Auto-Sklearn — what it does & limits

Materials: <https://ml-course.github.io/master/notebooks/06%20-%20Data%20Preprocessing.html>
(ML Engineering course — covers pipelines, SMOTE, preprocessing in depth)

Session 27

Topic: Exp 4.1

Materials: —

Session 28

Topic: Real-world data cleaning challenges & strategies

Materials: <https://www.kaggle.com/learn/data-cleaning> (Kaggle's free data cleaning course)

Session 29

Topic: Open dataset walkthrough: Kaggle/UCI exploration demo

Materials: <https://archive.ics.uci.edu/datasets> (UCI ML Repository — datasets for exploration)

Session 30

Topic: Exp 4.2 + 4.3

Materials: —

Session 31

Topic: Group project briefing + EDA planning strategies

Materials: <https://pycaret.org/tutorials/> (PyCaret AutoML tutorials)

Session 32

Topic: Instructor-guided full EDA demo on a new dataset

Materials: https://github.com/drshahizan/Python_EDA (End-to-end EDA reference with notebooks)

Session 33

Topic: Exp 4.4 + 4.5

Materials: —

Session 34

Topic: Time series preprocessing: resampling, smoothing, lag features

Materials: https://pandas.pydata.org/docs/user_guide/timeseries.html (Pandas time series — resampling, rolling mean, lag features)

Session 35

Topic: Text preprocessing: tokenisation, stop-words, stemming, lemmatisation

Materials: <https://www.analyticsvidhya.com/blog/2021/06/text-preprocessing-in-nlp-with-python-codes/> (Text preprocessing: tokenization, stop-words, stemming, lemmatization — NLTK & spaCy) https://www.slideshare.net/ankit_ppt/nlp-toolkits-andpreprocessingtechniques (NLP toolkits & preprocessing slides)

Session 36

Topic: Exp 5.1 + 5.2

Materials: —

Session 37

Topic: Imbalanced data: SMOTE, ADASYN, over- & under-sampling

Materials:

https://imbalanced-learn.org/stable/references/generated/imblearn.over_sampling.SMOTE.html (Official SMOTE & ADASYN docs)

Session 38

Topic: Dask for large-scale data; data augmentation for text & images

Materials: <https://docs.dask.org/en/stable/dataframe.html> (Dask DataFrame for large-scale data) <https://ml-course.github.io/master/notebooks/06%20-%20Data%20Preprocessing.html> (Same ML Engineering course — imbalanced data section)

Session 39

Topic: Exp 5.3 + 5.4 + 5.5

Materials: —

Session 40

Topic: Data versioning and pipeline monitoring: managing preprocessing workflows in production

Materials: https://github.com/smusab9152/Data_Preprocessing_and_Exploratory_data_analysis (End-to-end preprocessing + EDA notebooks — good capstone reference)

Session 41

Topic: Data drift detection with Evidently AI; writing modular, reusable preprocessing functions with live demo

Materials: <https://www.kaggle.com/learn/intro-to-machine-learning> (Kaggle's structured ML intro — good revision resource)

Session 42

Topic: Build a robust pipeline that handles unseen categories, mixed dtypes, and missing columns on a new dataset and log a data quality report

Materials: —

Session No.	Topics Covered	Learning Outcome	Activities / Tools Used
Module 1: Preprocessing Basics			
1	Raw data formats; data cleaning intro	Understand different raw data formats and data quality issues	Slides, Kaggle Pandas tutorial
2	Missing data: imputation methods (mean, median, mode) with Pandas demo	Apply imputation strategies to handle missing values	Pandas, scikit-learn impute docs, Jupyter Notebook
3	Exp 1.1 – Load CSV/JSON, print shape, dtypes & missing value counts; Exp 1.2 – Fill missing values using mean, median, mode and compare results	Load and inspect real datasets; compare imputation outcomes	Python, Pandas, Jupyter Notebook
4	Categorical encoding: one-hot & label encoding	Encode categorical features using appropriate techniques	Slides, scikit-learn preprocessing docs, Jupyter Notebook
5	Outlier detection: Z-score & IQR theory + visualisation	Detect and interpret outliers using statistical methods	Matplotlib, Seaborn, Analytics Vidhya reference
6	Exp 1.3 – Encode categorical column using label & one-hot encoding; Exp 1.4 – Detect and remove outliers using Z-score and IQR, compare row counts	Implement encoding and outlier removal on real datasets	Python, Pandas, scikit-learn, Jupyter Notebook
7	Scaling & normalisation: Min-Max vs StandardScaler	Differentiate and apply scaling techniques appropriately	Slides, scikit-learn preprocessing docs
8	Module 1 review, Q&A, common real-world preprocessing pitfalls	Consolidate preprocessing concepts and identify common mistakes	Discussion, Q&A, whiteboard
9	Exp 1.5 – Apply Min-Max normalization and StandardScaler; plot before/after distributions	Compare effect of scaling techniques on feature distributions	Python, scikit-learn, Matplotlib, Jupyter Notebook
Module 2: Feature Engineering			
10	Feature selection: correlation methods overview	Understand and apply correlation-based feature selection	Slides, scikit-learn feature selection docs
11	Statistical tests for feature selection: chi ² , ANOVA, mutual info	Select relevant features using statistical significance tests	Slides, scikit-learn feature selection docs, Jupyter Notebook
12	Exp 2.1 – Compute correlation matrix and drop high-correlation features; Exp 2.2 – Use chi ² and ANOVA to select top 5 features	Implement and evaluate feature selection on classification datasets	Python, Pandas, scikit-learn, Seaborn, Jupyter Notebook
13	Dimensionality reduction: PCA theory & intuition	Explain PCA and its role in reducing feature space	Slides, Stanford CS229 PCA notes, scikit-learn decomposition docs
14	t-SNE: intuition, hyperparameters & use cases;	Distinguish t-SNE from PCA and	Slides, scikit-learn manifold

Sessio n No.	Topics Covered	Learning Outcome	Activities / Tools Used
	compare with PCA	apply it to high-dimensional data	docs, Jupyter Notebook
15	Exp 2.3 – Apply PCA to reduce dataset to 2 components and plot explained variance; Exp 2.4 – Apply t-SNE and visualise 2D output by class label	Reduce dimensionality and interpret visual outputs	Python, scikit-learn, Matplotlib, Jupyter Notebook
16	Feature creation: polynomial features & interaction terms	Generate new features to improve model expressiveness	Slides, scikit-learn polynomial features docs
17	Feature importance evaluation: RF-based and permutation importance	Evaluate and rank feature importance using model-based methods	Slides, scikit-learn permutation importance docs
18	Exp 2.5 – Generate polynomial features (degree 2) and interaction terms for a regression dataset; check new shape	Create and verify engineered features for a regression task	Python, scikit-learn, Pandas, Jupyter Notebook
Module 3: Data Visualization			
19	Matplotlib: line, bar, scatter — layout & styling principles	Create well-labelled static plots using Matplotlib	Slides, Matplotlib official tutorials, Jupyter Notebook
20	Seaborn: histograms, box plots, pair plots; when to use each	Select appropriate Seaborn plots for distribution analysis	Slides, Seaborn official tutorial, Jupyter Notebook
21	Exp 3.1 – Line, bar & scatter plots with Matplotlib; Exp 3.2 – Histogram and box plot with Seaborn to compare distributions	Build and interpret plots for datasets using Matplotlib and Seaborn	Python, Matplotlib, Seaborn, Jupyter Notebook
22	Heatmaps & correlation visualisation for EDA	Visualise feature relationships using heatmaps for EDA insights	Seaborn heatmap docs, Plotly Express, Jupyter Notebook
23	Plotly: interactive dashboards; extracting decisions from visuals	Build interactive visualisations and derive data-driven insights	Plotly Express official docs, Jupyter Notebook
24	Exp 3.3 – Correlation heatmap highlighting top pairs; Exp 3.4 – Interactive scatter with dropdowns using Plotly; Exp 3.5 – Dashboard-style multi-plot layout	Design and deploy interactive and multi-panel EDA visualisations	Python, Seaborn, Plotly, Matplotlib, Jupyter Notebook
Module 4: Case Studies & Tools			
25	Scikit-learn pipelines: design, structure, transforms	Design modular preprocessing pipelines using scikit-learn	Slides, scikit-learn Pipeline & ColumnTransformer docs
26	AutoML intro: TPOT, Auto-Sklearn — what it does & limits	Understand AutoML tools and their role in automating model selection	Slides, ML Engineering course reference
27	Exp 4.1 – Build a scikit-learn pipeline with imputation, encoding, and scaling; fit on training set	Implement a complete sklearn pipeline on a real dataset	Python, scikit-learn, Pandas, Jupyter Notebook
28	Real-world data cleaning challenges & strategies	Identify and resolve common data quality issues in real datasets	Slides, Kaggle Data Cleaning course
29	Open dataset walkthrough: Kaggle/UCI exploration demo	Explore and profile an open dataset end-to-end	UCI ML Repository, Kaggle, Pandas, ydata-profiling
30	Exp 4.2 – Clean a real-world messy dataset (duplicates, inconsistent formats, missing values); Exp 4.3 – Profile a Kaggle/UCI dataset using ydata-profiling and summarise findings	Apply cleaning strategies and automated profiling to real datasets	Python, Pandas, ydata-profiling, Jupyter Notebook
31	Group project briefing + EDA planning strategies	Plan and scope an end-to-end EDA project as a team	PyCaret tutorials, group discussion

Session No.	Topics Covered	Learning Outcome	Activities / Tools Used
32	Instructor-guided full EDA demo on a new dataset	Observe and replicate a structured EDA workflow	Python, Pandas, Matplotlib, Seaborn, GitHub EDA reference
33	Exp 4.4 – Run AutoML using PyCaret and compare top 3 models; Exp 4.5 – End-to-end group EDA covering shape, distributions, nulls, correlations, and insights	Evaluate AutoML outputs and conduct a collaborative EDA	Python, PyCaret, Pandas, Seaborn, Jupyter Notebook
Module 5: Advanced Data Handling			
34	Time series preprocessing: resampling, smoothing, lag features	Preprocess time-series data using resampling and rolling statistics	Slides, Pandas time series docs
35	Text preprocessing: tokenisation, stop-words, stemming, lemmatisation	Apply NLP preprocessing techniques to raw text data	Slides, Analytics Vidhya NLP guide, NLTK, spaCy
36	Exp 5.1 – Resample time-series from daily to weekly and apply rolling mean; Exp 5.2 – Clean text column: tokenise, remove stop-words, stem and lemmatise	Process time-series and text data using appropriate Python libraries	Python, Pandas, NLTK, spaCy, Jupyter Notebook
37	Imbalanced data: SMOTE, ADASYN, over- & under-sampling	Handle class imbalance using sampling techniques	Slides, imbalanced-learn official docs
38	Dask for large-scale data; data augmentation for text & images	Apply Dask for scalable preprocessing and augmentation for small datasets	Slides, Dask DataFrame docs, ML Engineering course reference
39	Exp 5.3 – Handle imbalanced dataset using SMOTE and ADASYN; compare class distributions; Exp 5.4 – Augment text dataset using synonym replacement; Exp 5.5 – Build automated preprocessing pipeline using Pandas and Dask	Balance datasets and build scalable preprocessing pipelines	Python, imbalanced-learn, NLTK, Dask, Pandas, Jupyter Notebook
40	Data versioning and pipeline monitoring: managing preprocessing workflows in production	Understand production-level data pipeline management practices	Slides, GitHub EDA reference notebooks
41	Data drift detection with Evidently AI; writing modular, reusable preprocessing functions with live demo	Detect data drift and design reusable preprocessing code	Evidently AI, Python, Jupyter Notebook, Kaggle ML intro
42	Build a robust pipeline handling unseen categories, mixed dtypes, and missing columns on a new dataset; log a data quality report	Construct a production-ready, fault-tolerant preprocessing pipeline	Python, scikit-learn, Pandas, Jupyter Notebook